Macroeconomics I — MPE Insper

Practice list: Measurement, Welfare and Growth — qualifier level (Listas 1–2)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — Price indices, new goods, welfare and growth accounting

(Kurlat chs. 1–2, 5 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) A household has CES preferences with elasticity of substitution $\eta = 3$ between two goods and, at base prices, spends half of its budget on each. The price of good 1 rises 50% and the price of good 2 does not change. By how much does a Laspeyres index overstate the true change in the cost of living?

A. 2.1%

B. 6.2%

C. 25.0%

D. 0%

b) (3 points) *Welfare × growth.* Over 30 years Brazil's mean consumption grows 1.5% a year, life expectancy rises from 66 to 76 years, the variance of log consumption falls from 1.2 to 1.0, and leisure does not change. With the Kurlat–Jones–Klenow utility, $\sigma = 1$, $\bar{u} = 5$ and initial mean consumption normalized to one, the average annual growth of consumption-equivalent welfare is:

A. 1.50%

B. 1.83%

C. 4.33%

D. 2.30%

c) (2 points, T or F) A Paasche price index can never exceed the corresponding Laspeyres index.

d) (2 points, T or F) *PPP × productivity growth.* Tradables obey the law of one price and the nominal exchange rate is constant. If Brazil's tradable-sector productivity grows 2 points a year faster than the US's while non-tradable productivity grows equally in both, Brazil's price level relative to the US rises by 2γ points a year, γ being the non-tradable share in consumption.

Question 2. (15 points) Discursive. *Measurement × growth accounting.* Consumers have CES preferences over the set Ω_t of varieties available at date t,

$$U = \Big(\sum_{i \in \Omega_t} (b_i x_i)^{\frac{\eta-1}{\eta}} \Big)^{\frac{\eta}{\eta-1}}, \qquad \eta > 1,$$

where $b_i > 0$ is a quality weight. The statistical agency observes prices and expenditure shares and builds its deflators from varieties present in both periods.

a) (3 points) Find the minimum expenditure needed to obtain one unit of utility at prices p, and each variety's expenditure share.

Solution:

b) (3 points) Keep the set of varieties fixed between dates 0 and 1. Express the exact index P_1/P_0 in terms of observable price relatives $R_i = p_{i1}/p_{i0}$ and shares, rank it against the Laspeyres and Paasche indices, and approximate the Laspeyres bias to second order in the dispersion of $\ln R_i$.

Solution:

c) (3 points) Now some varieties are new at date 1 and none disappears. Find the exact index in terms of the index over continuing varieties and observable shares. What happens as $\eta \to \infty$ and as $\eta \to 1$?

Solution:

d) (3 points) Continuing-variety consumption inflation is 5% and new varieties take 4% of date-1 consumption spending; $\eta = 4$. New varieties appear only among consumption goods (share $s_C = 0.8$ of GDP); investment goods (share $s_I = 0.2$) have their own deflator bias b_I from unmeasured quality change. In a steady state with $\alpha = 1/3$, how is the Solow residual of growth accounting distorted? Compute it for $b_I = 0$, $b_I = b_C$ and $b_I = 3b_C$, where b_C is the consumption-deflator bias.

Solution:

e) (3 points) *Parameter analysis.* How do the inflation bias and the residual bias depend on η, the new-variety share $\nu \equiv 1 - \lambda_1$ and b_I? Prove the signs, fill the inflation bias for $\eta \in \{2, 4, 8\}$ and $\nu \in \{2\%, 4\%, 8\%\}$, find the b_I at which the residual bias changes sign, and draw the inflation bias against η for the three values of ν.

Solution:

Block 2 — Growth, capital flows and national income

(Kurlat chs. 4–5; Romer chs. 1, 4 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) *Development accounting × capital flows.* Let $Y = AK^{\alpha}(hL)^{1-\alpha}$ with $\alpha = 0.4$ and the same A in Brazil and the US. Brazil's output per worker is 25% of the US's and its human capital per worker 60%. The ratio of marginal products of capital (Brazil/US) implied by the closed-economy model is about:

A. 8.0

B. 3.7

C. 1.0

D. 2.5

b) (3 points) With $f(k) = [\alpha k^{-1} + 1 - \alpha]^{-1}$ (elasticity of substitution 1/2), $\alpha = 1/2$ and $n + g + \delta = 0.08$, the Golden Rule saving rate is:

A. 50%

B. 20%

C. 8%

D. 32%

c) (2 points, T or F) With CES production and an elasticity of substitution different from one, a balanced growth path with a constant capital–output ratio exists only if technical progress is labor-augmenting.

d) (2 points, T or F) *Growth × open economy.* In a small open Solow economy with perfect capital mobility, a permanent rise in the saving rate permanently raises GDP per worker.

Question 4. (15 points) Discursive. *Growth × national accounts × capital flows.* Brazil as a small open economy. Output is $Y = K^{\alpha}(AhL)^{1-\alpha}$; capital moves freely and earns the world rate r^* net of depreciation δ; A grows at g, L at n, h is constant. Residents own wealth W (capital at home or abroad) and save a fraction s of gross national income (wages plus the gross return on wealth): $\dot{W} = s\,\mathrm{GNI} - \delta W$. Numbers: $\alpha = 1/3$, $r^* = 0.04$, $\delta = 0.05$, $n + g = 0.03$.

a) (3 points) Characterize GDP per effective worker and its dynamics. What does a rise in s do to GDP? What does this imply for the Lucas paradox?

Solution:

b) (3 points) Derive the law of motion of wealth per effective worker, its steady state and the condition for stability.

Solution:

c) (3 points) Find the saving rate above which Brazil is a net creditor in the steady state and interpret it. Find GNI/GDP in the steady state in closed form.

Solution:

d) (3 points) With $s = 0.15$ compute k^*, y^*, ω^*, net foreign assets, the share of domestic capital owned abroad, GNI/GDP and the half-life of wealth. Draw $\dot{\omega}$ against ω for $s = 0.15$ and $s = 0.25$, marking k^*, and describe the transition after s rises.

Solution:

e) (3 points) *Parameter analysis.* How do ω^*/k^*, GNI/GDP and the half-life of wealth respond to s, r^*, $n+g$ and α? Prove the signs, fill the table for the baseline and for $s = 0.25$, $r^* = 0.06$, $n+g = 0.02$, $\alpha = 0.4$ (one change at a time), and give the limits as $s \to 0$ and as s approaches its upper bound.

Solution: